



Associative Classifier Applied to Medical Data

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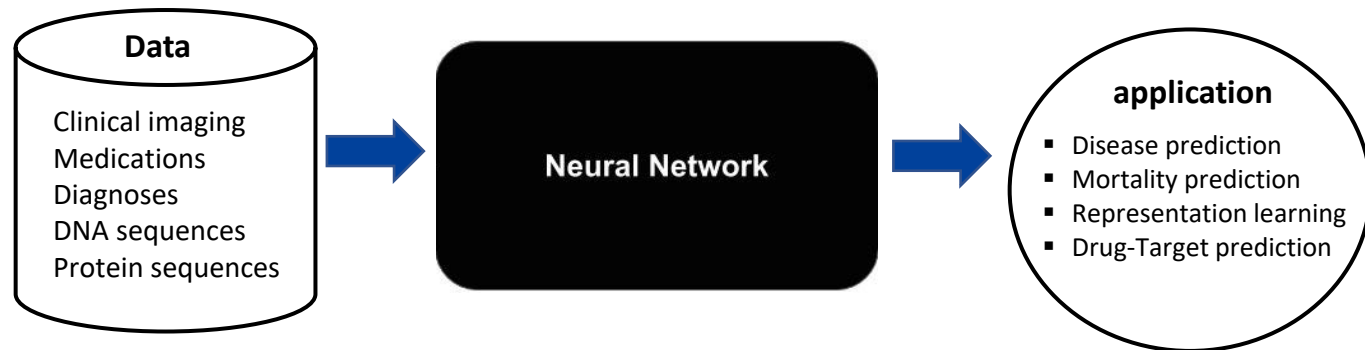
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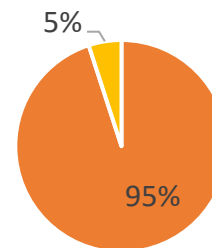
PROBLEM STATEMENT



- ❖ ML as a powerful tool in healthcare and EMR analysis.
- ❖ Prevalent techniques, like ANNs and SVMs behave like a black box: lack interpretability.
- ❖ Class imbalance problem is common in healthcare.
- ❖ Traditional ML models favor majority class -> poor performance on minority class



Distribution of Healthy and Diseased Patients



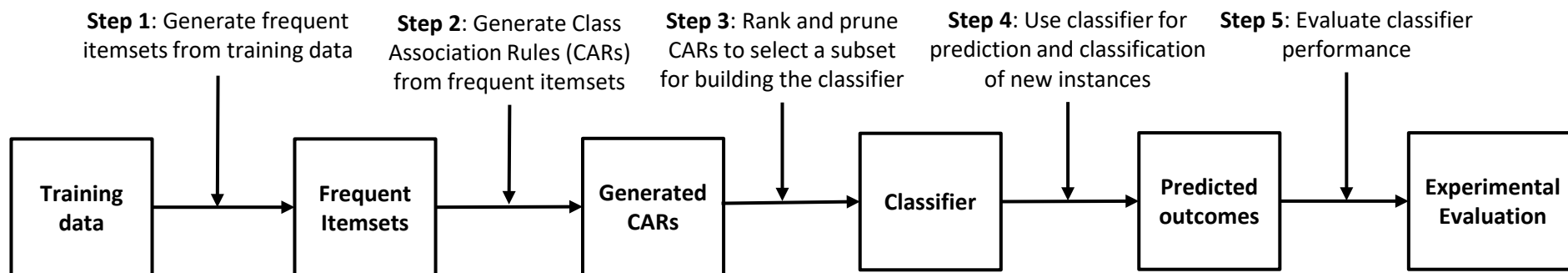
- Healthy Patients
- Patients with Disease

ASSOCIATIVE CLASSIFIERS



- ❖ A supervised learning model that generates association rules for classification
- ❖ Known for their high interpretability and readability
- ❖ Generate if-then rules in the form {set_of_features → class_label}
- ❖ Notable examples include CBA, CPAR, CMAR, and CCS

General steps in Associative Classification



PROPOSED METHOD – RULE GENERATION

- ❖ Predicts patient mortality based on drugs administered during hospital stay
- ❖ Maps class labels to 0 (survival) and 1 (mortality)
- ❖ Generates rules based on interestingness measure $I(i)$

$$I(i) = \begin{cases} 1 - \frac{sup_1(i)}{sup_0(i)} & \text{for rule } i \rightarrow 0 \\ 1 - \frac{sup_0(i)}{sup_1(i)} & \text{for rule } i \rightarrow 1 \end{cases} \quad \text{where } sup_c(x) = \frac{\sigma(x,c)}{\sigma(c)}$$

Rule Generation (training set T, minSupport, k)

1. Let C be the set of class labels 0, 1
2. Rules $\leftarrow \emptyset$
3. Apply the Apriori algorithm on T to find frequent itemsets (F) with minSupport
4. for each $i \in F$
5. if $sup_0(i) \geq sup_1(i)$
6. score = $1 - sup_1(i) / sup_0(i)$
7. add rule $i \rightarrow 0$ with score to Rules
8. else
9. score = $1 - sup_0(i) / sup_1(i)$
10. add rule $i \rightarrow 1$ with score to Rules
11. Return Rules

RULE PRUNING



- ❖ **Rules are sorted** in descending order by their scores
- ❖ The **top 'k' rules** that cover each transaction for each class are identified
- ❖ The strength of each rule is estimated based on its **predictive accuracy (F1 score)**
- ❖ Accumulation of rule strengths helps identify a **cut-off point** for pruning.
- ❖ Rules beyond this point are **pruned**, and the rest are used in classification

Rule Pruning (Rules)

1. Sort rules in Rules in decreasing order by interestingness score
2. for each rule r in sorted order
3. Estimate the strength of the rule r based on the prediction on the training set
4. Accumulate the rules strength and find maximum value to keep the rules until that point
5. Return the selected rules

CLASSIFICATION



- ❖ Predicting patient mortality for each instance in the test set
- ❖ For each rule that covers the instance, a score is computed for its class
- ❖ Instances are classified as the class with the maximum score

Classification (object o, SelectedRules, k)

1. $S \leftarrow [0, 0]$
2. counter0 = counter1 = 0
3. for each r in SelectedRules
4. if r covers o
5. if r.class == 0 and counter0 < k
6. counter0++
7. $S[0] = S[0] + \text{rule.interestingness_score}$
8. else if r.class == 1 and counter1 < k
9. counter1++
10. $S[1] = S[1] + \text{rule.interestingness_score}$
11. predClass = $\text{argmax}_{c \in C} S[c]$
12. return predClass

EXPERIMENTAL SETUP



- ❖ Tested on a dataset obtained from one hospital in the *eICU database*
- ❖ Significant class imbalance: majority class 0 occupies 90.6%
- ❖ The dataset was split with a training-test ratio of 9:1

Dataset summary

Drug features	Total patients	Survived patients (class 0)	Deceased patients (class 1)
147	4615	4179	436

Classifier Parameters

Classifier	Parameters
CBA	Min support: 1%, Min confidence: 50%
CMAR	Min support: 1%, Min confidence: 50%
Decision Tree	Quality Measure: Entropy
Proposed Method	Min support: 2%, k: 3

EXPERIMENTAL RESULTS



- ❖ Proposed classifier outperformed the other methods
- ❖ 80% of the 1254 rules were pruned, leaving only 251 rules
- ❖ Produces rules such as {*ACETAMINOPHEN*, *GLUCOSE*} → *survived*, indicating likely survival if both drugs are administered.
- ❖ These interpretable rules enhance transparency, aid understanding of survival factors, and assist medical professionals in optimizing treatment plans.

Performance comparison

Classifiers	F1	ROC AUC
Proposed classifier	0.5396	0.7553
CBA	0.4915	-
Decision Tree	0.4921	0.7286
CMAR	0.0000	0.5000

CONCLUSIONS AND FUTURE WORK



Conclusions

- ❖ Novel associative classifier specifically designed to handle unbalanced healthcare data
- ❖ Rule pruning strategy effectively reduces the rule set, enhancing interpretability
- ❖ Model produces readable and interpretable rules
- ❖ Superior performance on a highly unbalanced hospital dataset

Future Work

- ❖ Explore performance on aggregated data from multiple hospitals to evaluate its generalizability and robustness